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Repository User Guide

Paper: Energetic Patterns of Cyclones in the Southwestern Atlantic

Auto-generated from repository READMEs

Section: Project Overview

Source: README.md

Energetic Patterns of Cyclones in the Southwestern Atlantic

This repository organises all scripts, data, and results for the paper on energetic patterns of South Atlantic extratropical cyclones, based on Chapter 6 of the PhD thesis. Cyclones are classified into three Energy Patterns (EP1, EP2, EP3) via PCA-based K-Means clustering of Lorenz Energy Cycle diagnostics during the intensification phase. The **current scientific focus** is `scripts/ep_structure_analysis/`, which performs composite analysis of ERA5 reanalysis fields to characterise the atmospheric structure of EP1 (N=444) and EP2 (N=979) cyclones during intensification.

1. Setup Environment

```
# Clone the repository
git clone https://github.com/daniloceano/paper_energy_patterns.git
cd paper_energy_patterns

# Create conda environment and install dependencies
bash setup_environment.sh

# Activate environment
conda activate paper_energy_patterns
```

2. Preprocess Data (Run Once)

```
# Download and cache all energy data for fast access
python scripts/preprocess_data/run_all.py
```

Expected outputs: data/tracks_SAt_filtered_with_energetics_processed.csv, data/energy_cache.parquet

3. Run Cluster Analysis (Energy Patterns)

```
python scripts/cluster_analysis_energy_patterns/run_all.py
```

Outputs clustering results to results/cluster/ and figures to figures/.

4. Generate Final Paper Figures

```
python scripts/main/run_all.py
```

Outputs publication-ready figures to figures/main/.

```
.
data/                                # Input and processed data
energy_cache.parquet                 # Preprocessed energy data (generated)
era5_ep_structure/                   # ERA5 composites for ep_structure_analysis
precomputed_composites_ep1.nc
precomputed_composites_ep2.nc
README.md
docs/                                # Generated documentation PDFs
scientific_notes_cluster_analysis.pdf
scientific_notes_ep_structure.pdf
user_guide_repository_readmes.pdf   # Auto-generated (see below)
figures/                             # Generated figures
exploratory/                         # Exploratory figures
main/                               # Final publication figures
results/                            # Analysis results
```

```

cluster/                # Cluster assignments and models
scripts/                # All analysis scripts
cluster_analysis_energy_patterns/ # PCA + K-Means clustering pipeline
ck_subterms_analysis/   # Barotropic conversion (Ck) decomposition
documentation/          # Compile READMEs into a PDF user guide
ep_structure_analysis CURRENT FOCUS: ERA5 composite analysis/ #
exploratory/            # Preliminary exploratory scripts
main/                   # Final publication figure scripts
preprocess_data/        # Data download and preprocessing
setup_and_examples/     # Environment verification and templates
utils/                  # Shared utility functions
activate.sh              # Quick environment activation
requirements.txt
setup_environment.sh

```

See `scripts/README.md` for detailed information on each subdirectory.

Cyclone Tracks and Energetics

- **DOI:** 10.5281/zenodo.18133432
- **Description:** Combined cyclone tracks and semi-Lagrangian Lorenz Energy Cycle diagnostics (2020, ~6,700 cyclones, 42 years)1979
- **Access:** Downloaded automatically by `scripts/preprocess_data/extract_tracks_from_zenodo.py`

LEC Results with Vertical Resolution

- **DOI:** 10.5281/zenodo.18243447
 - **Description:** Complete LEC results with vertical resolution (~1,500 cyclones, 32 pressure levels, 3-hourly)
 - **Access:** Downloaded automatically by `scripts/preprocess_data/download_lec_from_zenodo.py`
-

echo Configuration###

All scripts use a header-level configuration no command-line arguments needed:block

```

# Example configuration (top of any script)
SAMPLE_SIZE = 0          # Number of cyclones (0 = all)
USE_PARALLEL = True      # Enable parallel processing
N_WORKERS = 8            # Workers (adjust for your system)
DPI = 300                # Figure output quality

```

1. Energy Pattern Classification (`scripts/cluster_analysis_energy_patterns/`)

Normalises LEC energy variables, applies PCA by lifecycle phase, determines the optimal number of clusters via Gap Statistic, and classifies each cyclone into one of three Energy Patterns using K-Means. Key results: EP1 (11.6%, 444 high conversions; EP2 (25.6%, 979 moderate conversions; EP3 (62. weak/background energetics.7%) cyclones)

2 Spatial Structure Analysis (`scripts/ep_structure_ *Current Focus*analysis/`) .

Composite ERA5 analysis of EP1 and EP2 cyclones during intensification. Uses a storm-centred domain at 0.resolution to compute EGR (850 hPa), PV (200/850 hPa), temperature advection (850 hPa), moisture flux divergence (975 hPa), SLP, RK criterion (250 hPa), KE advection (250 hPa), and AFC (250 hPa), plus anomalies relative to the 2020 WMO climatology. Data stored in `data/era5_ep_structure/.1991500253030`

3. Barotropic Conversion Decomposition (`scripts/ck_subterms_analysis/`)

Decomposes the barotropic conversion term (Ck) into its three subterms for EP1 cyclones, which present the largest barotropic conversions in the dataset.

4. Final Paper Figures (`scripts/main/`)

Scripts numbered 07 (main figures) and S3 (supplementary) generate publication-ready figures at 300 DPI according to Scientific Reports guidelines.S101

First Time

```
bash setup_environment.sh
```

The script creates the `paper_energy_patterns` conda environment (Python 3.13), installs all packages from `requirements.txt`, and verifies the installation.

Daily Use

```
source activate.sh
# or
conda activate paper_energy_patterns
```

Verify Installation

```
python scripts/setup_and_examples/verify_environment.py
```

A consolidated user guide (docs/user_guide_repository_readmes.pdf) is auto-generated from all repository READMEs. To regenerate it:

```
python scripts/documentation/compile_docs.py
```

Requires pandoc and pdflatex on your system.

Section: Data

Source: data/README.md

Data Directory

Input Data (Accessed from Zenodo)

Input data **does not need to be downloaded** - accessed directly via URL:

Cyclone Tracks and Energetics (Integrated Dataset)

- **DOI:** 10.5281/zenodo.18133432
- **Direct URL:** https://zenodo.org/records/18133432/files/tracks_SAt_filtered_with_energetics.csv
- **Citation:** de Souza, D., & Gramscianinov, C. (2026). Southwestern Atlantic Cyclone Tracks and Semi-Lagrangian Lorenz Energy Cycle (LEC) diagnostics (1979–2020) [Data set]. Zenodo.
- **Description:** Complete dataset combining cyclone tracks with semi-Lagrangian Lorenz Energy Cycle diagnostics
- **Period:** 1979-2020 (42 years)
- **Format:** Long format CSV (each row = one time step of one cyclone)
- **Size:** 180.8 MB
- **Temporal Resolution:**
 - Track variables (position, vorticity): 1-hourly
 - Energy variables (LEC terms): 3-hourly

Main Columns: Identification and Location: - **track_id:** Unique cyclone identifier (format: YYYYNNNN) - **date:** UTC timestamp - **lon vor, lat vor:** Cyclone center coordinates (degrees) - **vor42:** Filtered 850 hPa vorticity ($\times 10^{-1}$, positive values) - **region:** Genesis region (ARG, LA-PLATA, SE-BR) - **period:** Life cycle phase (incipient, intensification, mature, decay)

Energy Reservoirs (J m^{-2}): - Az, Ae, Kz, Ke

Conversion Terms (W m^{-2}): - Cz, Ca, Ce, Ck

Boundary Terms (W m^{-2}): - BAz, BAe, BKz, BKe (flux terms) - B Φ Z, B Φ E (pressure-work terms)

Generation Terms (W m^{-2}): - Gz, Ge (diabatic generation)

Tendencies (W m^{-2}): - Az/ t, Ae/ t, Kz/ t, Ke/ t

Residuals (W m^{-2}): - RGz, RGe, RKz, RKe

Key Features:

- **Semi-Lagrangian:** $15^{\circ} \times 15^{\circ}$ storm-following control volume
- **Quality Controlled:** Minimum lifetime/displacement thresholds applied
- **Phase Attribution:** Objective life-cycle segmentation using CycloPhaser
- **Missing Values:** Some timestamps may contain NaN for energy terms

Processed Data (This directory)

Use this folder to save: - Intermediate processed data - Aggregations and statistics - Data subsets for specific analyses - Classification/clustering results

Note: Don't version very large files. Use `.gitignore` as needed.

Section: Scripts

Source: `scripts/README.md`

Scripts Directory

This directory contains all analysis scripts organised by purpose. The subdirectories range from preprocessing and utility functions to the final publication figure scripts and the current scientific analysis pipeline.

Directory Structure

```
scripts/
main Final publication figure scripts (07, S3)S101/          #
exploratory/          # Preliminary exploratory scripts (not in paper)
cluster_analysis_energy_patterns/ # PCA + K-Means Energy Pattern classification
ep_structure_analysis CURRENT FOCUS: ERA5 composite analysis (EP1 vs EP2)/ #
ck_subterms_analysis/ # Ck decomposition into subterms for EP1
preprocess_data/      # Data download and preprocessing
utils/                # Shared utility functions
setup_and_examples/   # Environment verification and script templates
documentation/        # Compile all READMEs into a consolidated PDF
```

Directory Descriptions

Final Publication Figure Scripts

Repository of final publication-ready figure scripts, numbered by figure order (07 main figures, S3 supplementary). Output goes to `figures/main/.S101`

Main scripts:

Script	Figure
01_figure_tracks_genesis_frequency.py	Fig 1: Study area and workflow
02_figure_20070643_publication.py	Fig 2: Case study cyclone 20070643
03_make_phase_density_2x2.py	Fig 3: Phase space density (22)
04_figure_lps_combined.py	Fig 4: Lorenz Phase Space for EP3
05_figure_intensity_seasonality_trends.py	Fig 5: EP intensity, seasonality, trends
06_figure_genesis_density_kde.py	Fig 6: Genesis density (KDE)
07_figure_ep1_instability_composite.py	Fig 7: EP1 instability composite
S1_figure_pca_clustering_validation.py	Fig S1: PCA/clustering validation
S2_figure_selected_tracks.py	Fig S2: Selected EP1 tracks
S3_figure_vertical_levels.py	Fig S3: Vertical energy conversion distribution
run_all.py	Run all figure scripts sequentially

Inputs: data/tracks_SAt_filtered_with_energetics_processed.csv, results/cluster/kmeans_clustered
S2 and S3 additionally require composites from data/era5_ep_structure/ produced by ep_structure_analysis.

Outputs: figures/main/

Preliminary Exploratory Scripts

General preliminary exploratory scripts. These predate or support the main pipeline but are not directly used in the final paper.

Scripts:

Script	Description
analyze_ep_characteristics.py	EP characteristic statistics
density_diagrams_with_ge.py	Density diagrams including Ge term
exploring_clustering.ipynb	Interactive clustering exploration
exploring_pca.ipynb	Interactive PCA exploration
figure_genesis_density_relative_kde.py	Relative genesis density
figure_minmax_vs_zscore_comparison.py	Normalisation method comparison
figure_most_intense_cyclone_lps.py	Most intense cyclone LPS
figure_three_intense_cyclones_individual.py	Three intense cyclones individual plots
figure_three_intense_cyclones_individual_zscore.py	Three intense cyclones individual zscore
kde_pairplot.py	KDE pairwise plot of energy terms
plot_ep_lps_diagrams.py	LPS diagrams per energy pattern
plot_pv_jet_composite.py	PV and jet composite
precompute_composites.py	Early composite precomputation prototype
scatter_density.py	Scatter density plots
vertical_term_boxplots_ep1_ep2.py	Vertical term boxplots EP1 vs EP2

Inputs: data/energy_cache.parquet, results/cluster/kmeans_clustered_data.csv

Outputs: figures/exploratory/

cluster_analysis_energy_ Energy Pattern Classificationpatterns/

Scripts for generating the Energy Patterns via PCA + K-Means clustering of Lorenz Energy Cycle diagnostics during the intensification phase.

Pipeline (run in order):

1. `step1_normalize_and_pca`. Normalise energy variables and apply PCA by lifecycle phasepy
2. `step2_plot_pca_results`. Visualise PCA loadings and explained variancepy
3. `step3_optimal_k_analysis`. Determine optimal k via Gap Statisticpy
4. `step4_apply_kmeans`. Apply K-Means (k=3) and assign Energy Patternspy
5. `step5_plot_energy_patterns`. Composite statistics and LPS diagramspy
6. `step6_generate_scientific_notes_pdf`. Convert SCIENTIFIC_NOTES.md to PDFpy

Inputs: `data/energy_cache.parquet`, `data/tracks_SAt_filtered_with_energetics_processed.csv`

Outputs: `results/cluster/` (CSV assignments, PCA/KMeans model pickles), `figures/`, `docs/scientific_notes_cluster_analysis.pdf`

Key results: EP1 (11.6%, N=444), EP2 (25.6%, N=979), EP3 (62.7%)

Current Scientific Focus)

Composite analysis of ERA5 reanalysis fields to understand the atmospheric structure of EP1 (N=444) and EP2 (N=979) cyclones during intensification. EP3 is excluded because it represents less intense, climatological-background cyclones.

Scientific Summary Objective: Investigate what structural differences in the large-scale atmospheric environment distinguish EP1 (high-conversion) from EP2 (moderate-conversion) cyclones.

Sample: All 444 EP1 and 979 EP2 cyclones identified by `cluster_analysis_energy_patterns`, using intensification-phase timesteps only.

ERA5 data: 0.resolution, storm-centred domain, 6-hourly. Pressure-level variables: u, v, t, z, q at 975 hPa. Single-level variable: msl.175303025

Diagnostics computed:

Dagnostic	Level(s)	Description
EGR (Eady Growth Rate)	850 hPa	Baroclinic instability measure
PV (Potential Vorticity)	200 hPa	Upper-level tropopause dynamics
PV	850 hPa	Low-level diabatic PV anomaly
Temperature Advection	850 hPa	Warm/cold advection patterns
Moisture Flux Divergence	975 hPa	Near-surface moisture convergence
SLP	Surface	Cyclone intensity and horizontal structure

Diagnostic	Level(s)	Description
RK criterion (Rayleigh-Kuo)	250 hPa	Barotropic/baroclinic instability condition
KE Advection	250 hPa	Jet-level kinetic energy tendency
AFC (Ageostrophic Flux Convergence)	250 hPa	Eddy KE redistribution

Anomaly versions (departure from 2020 WMO climatology) are computed for PV (200/850 hPa), temperature advection, moisture flux divergence, KE advection, and SLP.1991

Pipeline:

1. `step1_select_ep_tracks`. Select EP1/EP2 tracks from cluster results.py
2. `step2_download_era5_parallel`. Download ERA5 fields (run `**remotely**`)py
3. `step2_1_download_era5_monthly_means`. Download monthly mean climatology (run `**remotely**`)py
4. `step3_precompute_composites`. Compute composites (run `**remotely**`)py
5. `step4_create_figures`. Create composite figures (run `**locally**`)py
6. `step5_update_scientific_notes`. Update SCIENTIFIC_NOTES.md and regenerate PDF (run `**locally**`)py

Inputs: results/cluster/kmeans_clustered_data.csv, ERA5 via CDS API

Outputs: data/era5_ep_structure/precomputed_composites_ep1.nc, data/era5_ep_structure/precomputed_figures/, docs/scientific_notes_ep_structure.pdf

`ck_subterms_ Barotropic Conversion Decompositionanalysis/`

16.48).W m

Pipeline:

1. `step1_prepare_tracks`. Convert EP1 cyclone tracks to LorenzCycleToolkit format.py
2. `step2_run_lec_toolkit`. Run LorenzCycleToolkit with automatic ERA5 download.py
3. `step2_monitor_ck`. Monitor job progress.py

Prerequisite: Cluster results results/cluster/kmeans_clustered_data.csv (from cluster_analysis_energy)

Inputs: results/cluster/kmeans_clustered_data.csv, ERA5 (auto-downloaded by toolkit)

Outputs: data/ck_analysis/, results/ck_analysis/

`preprocess_ Data Download and Preprocessingdata/`

Scripts for downloading and preprocessing the input data from Zenodo.

Run order:

```
python scripts/preprocess_data/run_all.py
# or individually:
python scripts/preprocess_data/download_lec_from_zenodo.py
python scripts/preprocess_data/extract_tracks_from_zenodo.py
python scripts/preprocess_data/preprocess_data.py
```

Outputs: data/tracks_SAt_filtered_with_energetics_processed.csv, data/energy_cache.parquet, data/temp_lec_zenodo/

Shared Utility Functionsutils/

Shared utility functions used across the repository.

Module	Description
load_data.py	Load cyclone tracks and energy data
gap_statistic.py	Gap Statistic implementation (Tibshirani et al. 2001)

setup_and_ Environment Verification and Templatesexamples/

Script	Description
verify_environment.py	Check all required packages are installed
example_analysis.py	Minimal working example of a complete analysis
template_analysis.py	Boilerplate template for new scripts

Documentation Compilerdocumentation/

Contains compile_docs.py, which collects all repository READMEs and generates docs/user_guide_repository.

```
python scripts/documentation/compile_docs.py
```

How Imports Work

All scripts use absolute imports from the project root:

```
import sys
from pathlib import Path

project_root = Path(__file__).resolve().parents[2] # adjust depth as needed
if str(project_root) not in sys.path:
    sys.path.insert(0, str(project_root))

from scripts.utils.load_data import load_tracks
```

This ensures scripts work from **any working directory**: - From project root: `python scripts/main/01_figure_tracks_genesis_frequency.py` - From scripts dir: `python main/01_figure_tracks_genesis_frequency.py`

Depth reference: scripts in `scripts/` use `.parents[1]`; scripts in `scripts/subdir/` use `.parents[2]`.

Directory Setup

All scripts automatically create their output directories:

```
project_root = Path(__file__).resolve().parents[2]
FIGURES_DIR = project_root / "figures" / "my_analysis"
FIGURES_DIR.mkdir(parents=True, exist_ok=True)
```

Running Scripts

```
# From project root
python scripts/main/run_all.py
python scripts/cluster_analysis_energy_patterns/run_all.py

# From any subdirectory (imports resolve automatically)
cd scripts/main
python 01_figure_tracks_genesis_frequency.py
```

Available Utilities

```
from scripts.utils.load_data import (
    load_tracks,           # Load all cyclone tracks
    load_energy_by_cyclone, # Load energy for one cyclone
    load_all_energy_data,  # Load energy for multiple cyclones
)
from scripts.utils.gap_statistic import GapStatistic
```

Section: `scripts/main`

Source: `scripts/main/README.md`

Main Figure Scripts

This directory contains scripts for generating publication-ready figures used in the Energy Patterns manuscript. Each script produces figure(s) formatted according to Scientific Reports guidelines (300 DPI, specific dimensions, consistent styling).

Script Organization

Scripts are numbered according to the order of figures in the manuscript:

Main Figures

- `01_figure_tracks_genesis_frequency.py`
Figure 1: Study area and workflow overview
- `02_figure_20070643_publication.py`
Figure 2: Case study — Cyclone 20070643 energetics and trajectory
- `03_make_phase_density_2x2.py`
Figure 3: Phase space density distributions (2×2 layout)
- `04_figure_lps_combined.py`
Figure 4: Lorenz Phase Space for Energy Patterns (EP1–EP3)
- `05_figure_intensity_seasonality_trends.py`
Figure 5: Energy Pattern characteristics (intensity, seasonality, trends)
- `06_figure_genesis_density_kde.py`
Figure 6: Genesis density using Kernel Density Estimation (Hoskins & Hodges method)
- `07_figure_ep1_instability_composite.py`
Figure 7: EP1 instability composite (4×3 layout: RK, PV, EGR diagnostics)

Supplementary Figures

- `S1_figure_pca_clustering_validation.py`
Figure S1: PCA and clustering validation
- `S2_figure_selected_tracks.py`
Figure S2: Selected EP1 cyclones used in the spatial structure analysis
Requires: data/era5_ep_structure/precomputed_composites_ep1.nc (produced by scripts/ep_structure_analysis/step3_precompute_composites.py)
- `S3_figure_vertical_levels.py`
Figure S3: Vertical distribution of energy conversions for EP1 cyclones
Requires: data/era5_ep_structure/precomputed_composites_ep1.nc (produced by scripts/ep_structure_analysis/step3_precompute_composites.py)

Usage

Each script can be executed independently:

```
python scripts/main/01_figure_tracks_genesis_frequency.py
python scripts/main/02_figure_20070643_publication.py
# ... etc.
```

Or generate all figures sequentially:

```
python scripts/main/run_all.py
```

Dependencies

All figure scripts require: - Processed track data: `data/tracks_SAt_filtered_with_energetics_processed.csv`
- Clustering results: `results/cluster/kmeans_clustered_data.csv`

Some scripts have additional requirements (see individual script headers for details).

Outputs

Generated figures are saved to: `figures/main/`

Notes

- Figures S2 and S3 depend on outputs from the `ep_structure_analysis` pipeline (`scripts/ep_structure_analysis/`). Run steps 1–3 of that pipeline first to generate the composites in `data/era5_ep_structure/`.
- All scripts include configuration sections at the top for customizing plot parameters (colors, dimensions, labels, etc.) without modifying core logic.
- For detailed figure descriptions, methodology, and scientific interpretation, see `figures/main/README.md`.

Section: `scripts/exploratory`

Source: `scripts/exploratory/README.md`

Exploratory Scripts

General preliminary exploratory scripts used during the investigation phase of the project. These scripts are **not part of the final paper pipeline** — they predate or informally support the main analyses but are not called by any production pipeline.

Purpose

This directory holds scripts that were used to: - Explore the energy term distributions across Energy Patterns - Compare normalisation approaches before selecting the final method - Prototype composite calculations - Generate informal figures for internal scientific discussion

Scripts

Script	Description
<code>analyze_ep_characteristics.py</code>	Compute and tabulate statistical characteristics of each Energy Pattern
<code>density_diagrams_with_ge.py</code>	Density diagrams including the diabatic generation term Ge
<code>exploring_clustering.ipynb</code>	Interactive Jupyter notebook for K-Means clustering exploration

Script	Description
<code>exploring_pca.ipynb</code>	Interactive Jupyter notebook for PCA exploration
<code>figure_genesis_density_relative_kde.py</code>	Relative genesis density maps using KDE
<code>figure_minmax_vs_zscore_comparison.py</code>	Comparison of min-max vs z-score normalisation on clustering outcome
<code>figure_most_intense_cyclone_lps.py</code>	Lorenz Phase Space diagram for the single most intense cyclone
<code>figure_three_intense_cyclones_individual_lps.py</code>	Individual LPS diagrams for three intense cyclones
<code>figure_three_intense_cyclones_individual_zscore.py</code>	Individual z-score of the three intense cyclone plots
<code>kde_pairplot.py</code>	KDE pairwise contour plot matrix for selected energy terms across lifecycle phases
<code>plot_ep_lps_diagrams.py</code>	LPS diagrams grouped by Energy Pattern
<code>plot_pv_jet_composite.py</code>	Early prototype composite of PV and jet-stream fields
<code>precompute_composites.py</code>	Early prototype for precomputing ERA5 composites (superseded by <code>ep_structure_analysis/step3_precompute_composites.py</code>)
<code>scatter_density.py</code>	Scatter density plots of energy term pairs
<code>vertical_term_boxplots_ep1_ep2.py</code>	Box plots of vertical energy terms comparing EP1 and EP2

Inputs

- `data/energy_cache.parquet` — Preprocessed energy data (generated by `scripts/preprocess_data/preprocess_data.py`)
- `results/cluster/kmeans_clustered_data.csv` — Cluster assignments (generated by `scripts/cluster_analysis_energy_patterns/`)

Outputs

Figures are saved to `figures/exploratory/`. Results (if any) go to `results/exploratory/`.

Notes

- Scripts in this directory are standalone — they do not depend on each other.
- No `run_all.py` orchestration is provided; run individual scripts as needed.
- For the final paper figure scripts, see `scripts/main/`.

Section: `scripts/cluster_analysis_energy_patterns`

Source: `scripts/cluster_analysis_energy_patterns/README.md`

Cluster Analysis of Cyclone Energy Patterns

Objective: Identify and characterize distinct energetic patterns in South Atlantic cyclones through objective clustering analysis.

Scientific Documentation

Complete methodology, results, and interpretation:

→ SCIENTIFIC_NOTES.md

Generate PDF version:

```
python scripts/cluster_analysis_energy_patterns/generate_scientific_notes_pdf.py
```

Output: docs/scientific_notes_cluster_analysis.pdf

Requirements: - Pandoc: `brew install pandoc` (macOS) or `sudo apt install pandoc` (Linux)
- XeLaTeX: `brew install basictex` (macOS) or `sudo apt install texlive-xetex` (Linux)

Quick Summary

This analysis uses **PCA + K-Means clustering** to objectively classify South Atlantic cyclones into three distinct **Energy Patterns (EPs)** based on Lorenz Energy Cycle diagnostics.

Key Energy Terms: - **Ca**: Baroclinic conversion ($\text{APE} \rightarrow \text{eddy APE}$) - **Ck**: Barotropic conversion ($\text{KE} \rightarrow \text{eddy KE}$)

- **Ge**: Eddy APE generation - **BAe/BKe**: Boundary fluxes (energy import/export) - **Ae/Ke**: Energy reservoirs

Results: - **EP1 (11.6%)**: Strong barotropic and baroclinic conversions (mean $\text{Ck} = -16.48 \text{ W m}^{-2}$) - **EP2 (25.6%)**: Intermediate conversions (mean $\text{Ck} = -3.49 \text{ W m}^{-2}$) - **EP3 (62.7%)**: Weak energetics (mean $\text{Ck} = -1.71 \text{ W m}^{-2}$)

See SCIENTIFIC_NOTES.md for complete analysis.

Pipeline

Run All Steps

Complete pipeline (Steps 1-5)

```
python scripts/cluster_analysis_energy_patterns/run_all.py
```

Individual Steps

Step 1: Normalize and PCA

```
python scripts/cluster_analysis_energy_patterns/step1_normalize_and_pca.py
```

- Filters cyclones with complete lifecycles (3,820 cyclones, 56.3%)
- Phase-separated PCA (97% variance retained)

- Output: results/cluster_analysis_energy_patterns/pca_*

Step 2: Plot PCA Results

`python scripts/cluster_analysis_energy_patterns/step2_plot_pca_results.py`

- PC scatter plots, loadings heatmaps, scree plots
- Output: figures/cluster_analysis_energy_patterns/pca_*

Step 3: Optimal k Analysis

`python scripts/cluster_analysis_energy_patterns/step3_optimal_k_analysis.py`

- 5 cluster validity indices (Silhouette, Davies-Bouldin, Calinski-Harabasz, Score Function, Gap Statistic)
- Mean normalized score $\rightarrow$ **k = 3** optimal
- Output: figures/cluster_analysis_energy_patterns/optimal_k_*

Step 4: Apply K-Means

`python scripts/cluster_analysis_energy_patterns/step4_apply_kmeans.py`

- K-Means (k=3, n_init=100) on each phase
- Centroid reconstruction to original energy space
- Output: results/cluster_analysis_energy_patterns/kmeans_*

Step 5: Plot Energy Patterns

`python scripts/cluster_analysis_energy_patterns/step5_plot_energy_patterns.py`

- Lorenz Phase Space diagrams (Conversion + Imports)
- Zoom views for detailed pattern comparison
- Output: figures/cluster_analysis_energy_patterns/lps_*

Methodology Summary

Data Filtering

- **Original:** 6,789 cyclones
- **Filtered:** 3,820 cyclones (complete lifecycle: incipient $\rightarrow$ intensification $\rightarrow$ mature $\rightarrow$ decay)
- **Total records:** 15,280 ($3,820 \times 4$ phases)

Dimensionality Reduction

- **Input:** 7 energy terms (Ca, Ck, Ge, BAe, BKe, Ae, Ke)
- **Method:** Phase-separated PCA
- **Output:** ~6 PCs per phase (97% variance)

Clustering

- **Method:** K-Means (k=3, determined by 5 cluster validity indices)
- **Application:** Phase-separated clustering
- **Output:** 3 Energy Patterns (EP1, EP2, EP3)

Visualization

- **Lorenz Phase Space (LPS):**
 1. Conversion LPS: Ck (barotropic) vs Ca (baroclinic)
 2. Imports LPS: BAe (APE flux) vs BKe (KE flux)

See SCIENTIFIC_NOTES.md for detailed methodology - Quadrant analysis reveals conversion direction and efficiency - Marker characteristics add information about energy magnitude and generation

Why LPS instead of bar plots/heatmaps?: Energy terms have fundamentally different scales and physical meanings. The Lorenz Phase Space framework provides a physically meaningful visualization that respects the relationships between energy conversions and fluxes, rather than attempting direct magnitude comparisons across disparate variables.

Results Structure

Data Files (39 files)

PCA Results (20 files):

```
results/cluster/
  pca_scores_{phase}.csv          # PC coordinates (includes track_id)
  pca_full_data_{phase}.csv       # Original + scaled + PCs
  pca_models_{phase}.pkl         # PCA & StandardScaler objects
  pca_loadings_{phase}.csv       # Variable loadings on PCs
  pca_explained_variance_{phase}.csv # Variance explained by each PC
```

Optimal K Analysis (3 files):

```
results/cluster/
  optimal_k.txt                  # Single value: k=3
  optimal_k_raw_indices.csv      # Raw CVI values for k=3 to 15
  optimal_k_normalized_indices.csv # Normalized & averaged CVIs
```

Clustering Results (16 files):

```
results/cluster/
  kmeans_clustered_data_{phase}.csv # Original data + cluster labels
  kmeans_centroids_pc_{phase}.csv   # Centroids in PC space
  kmeans_centroids_energy_{phase}.csv # Centroids in energy space
  kmeans_model_{phase}.pkl          # KMeans model object
```

Results Structure

Data Files

```
results/cluster_analysis_energy_patterns/  pca_scores_{phase}.csv # PC scores for each cy-
clone  pca_loadings_{phase}.csv # Variable contributions to PCs  pca_variance_{phase}.csv
# Explained variance ratios  pca_models.pkl # Saved PCA transformations  opti-
mal_k_scores.csv # CVI scores for k=2..10  kmeans_clustered_data_{phase}.csv #
Cluster assignments  kmeans_centroids_{phase}.csv # Cluster centroids (PC space)
```

```
kmeans_centroids_energy_{phase}.csv # Centroids (original energy space)  kmeans_summary_{phase}.csv
# Cluster statistics
```

Figures

```
figures/cluster_analysis_energy_patterns/  pca/  scatter_{phase}.png # PC pair-wise plots
  loadings_{phase}.png # Variable loadings heatmaps  variance_{phase}.png # Scree plots
optimal_k_analysis.png # CVI comparison (k=2..10)  lps_conversion_zoom.png # Conversion
LPS (Ck vs Ca)  lps_imports_zoom.png # Imports LPS (BAe vs BKe)
```

References

Lorenz Energy Cycle:

- Lorenz, E. N. (1955). Available potential energy and the maintenance of the general circulation.

Cluster Validation:

- Rousseeuw, P. J. (1987). Silhouettes: a graphical aid to the interpretation and validation of cluster analysis.
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South Atlantic Cyclones:

- Reboita, M. S., et al. (2010). South Atlantic Ocean cyclogenesis climatology simulated by RegCM3.
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****Date:**** February 2026

****Last Updated:**** February 23, 2026

Section: scripts/ep_structure_analysis

> Source: `scripts/ep\_structure\_analysis/README.md`

EP Structure Analysis: EP1 vs EP2 Cyclone Comparison

Objective

Investigate the spatial structure of EP1 and EP2 cyclones during intensification, using standard dynamical diagnostics to understand what distinguishes each energy

pattern structurally.

Important: Consistency with Cluster Analysis

****CRITICAL:**** All cyclones used in this analysis come directly from the cluster assignments in `results/cluster/kmeans\_clustered\_data.csv`. The clustering was performed on cyclones already filtered for complete lifecycle (incipient → intensification → mature → decay). Therefore:

- ****NO additional lifecycle filtering**** is applied in this pipeline
- ****ALL 444 EP1 cyclones**** from cluster 0 are used (100% consistency)
- ****ALL 979 EP2 cyclones**** from cluster 2 are used (100% consistency)

This ensures that the structural analysis describes the ****exact same cyclones**** used to define the energy patterns, maintaining methodological consistency throughout the study.

Diagnostic Fields

Field	Levels	Purpose	Key References
EGR (Eady Growth Rate)	500-850 hPa layer	Measures baroclinic instability of the background	
PV (Potential Vorticity)	200 hPa	Upper-level tropopause dynamics and stratospheric influence	
PV (Potential Vorticity)	850 hPa	Low-level PV anomaly associated with surface cyclone	
Temperature advection	850 hPa	Warm/cold advection patterns linked to QG forcing for a	
Specific humidity	975 hPa	Low-level moisture distribution	Bao et al. (2002); Schär
Moisture flux divergence	975 hPa	Moisture convergence/divergence → convective potential	
SLP (Sea Level Pressure)	Surface	Cyclone position, intensity and horizontal structure	
RK criterion (Rayleigh-Kuo)	250 hPa	Barotropic/baroclinic instability necessary condition	
KE advection	250 hPa	Kinetic energy tendency from advection in jet stream	-
AFC (Ageostrophic Flux Convergence)	250 hPa	Eddy KE source/sink from ageostrophic pre	

****Anomaly diagnostics**** (departure from 1991-2020 WMO climatology - same decomposition as AFC)

Anomaly Field	Level	Output variable	Climatology group
PV	200 hPa	`pv_200_anom`	`pv200` (175/200/225 hPa, u,v,t)
PV	850 hPa	`pv_850_anom`	`pv850` (825/850/875 hPa, u,v,t)
Temp advection	850 hPa	`adv_T_850_anom`	`pv850` (u,v,t at 850 hPa)
Moisture flux div	975 hPa	`div_q_975_anom`	`mfd975` (975 hPa, u,v,q)
KE advection	250 hPa	`ke_adv_250_anom`	`250hPa` (u,v at 250 hPa)
SLP	Surface	`msl_anom`	`slp` (msl, single-level)

- > ****Note:**** EGR is not decomposed (layer-mean nature makes eddy decomposition ill-defined).
- > AFC is already an anomaly field by construction (uses ϕ and $\vec{v}$).

Level selection rationale

- ****EGR (250-850 hPa):**** The 250-850 hPa layer captures the main tropospheric baroclinic zone. This layer-mean approach is standard since Hoskins & Valdes (1990) and has been widely applied in Southern Hemisphere cyclone studies (Simmonds & Lim, 2009; Gramscianinov et al., 2019). The growth rate $= 0.31 \cdot f/N \cdot |V/z|$ integrates the static stability (N) and vertical wind shear over the full depth of the troposphere.
- ****PV at 200 hPa:**** The dynamic tropopause is traditionally defined as the 2 PVU surface, typically found near 200 hPa in midlatitudes (Hoskins et al., 1985). Upper-level PV anomalies (tropopause folds) are precursors for rapid cyclogenesis (Davis & Emanuel, 1991; Rossa et al., 2000). The 200 hPa level is widely used in extratropical cyclone composites (e.g., Dacre et al., 2012; Catto et al., 2010).
- ****PV at 850 hPa:**** Low-level PV anomalies are generated by diabatic heating (latent heat release) and are a key component of the "PV tower" structure in explosively deepening cyclones (Čampa & Wernli, 2012; Martínez-Alvarado et al., 2016). The 850 hPa level is standard for characterizing the lower-tropospheric cyclone circulation.
- ****Temperature advection at 850 hPa:**** The 850 hPa level is the standard reference for thermal advection in synoptic analysis (e.g., Sanders & Gyakum, 1980; Sinclair, 1994). Warm air advection (WAA) ahead of the surface cyclone is a primary quasi-geostrophic forcing for upward motion (Sutcliffe, 1947; Trenberth, 1978), while cold air advection (CAA) in the rear contributes to frontal structure and cyclone deepening.
- ****SLP:**** Standard field for cyclone tracking and composite analysis (Hoskins & Hodges, 2005; Reboita et al., 2010).
- ****Moisture fields at 975 hPa:**** The 975 hPa level captures near-surface moisture transport while remaining above the planetary boundary layer turbulence. Moisture flux divergence ($\nabla \cdot (qV)$) identifies regions of moisture convergence (negative values) associated with convective potential and latent heat release, a key diabatic process in cyclone intensification (Banacos & Schultz, 2005; Lackmann, 2011). The calculation uses MetPy's spherical geometry-aware gradient operators and physical constants to ensure consistency.
- ****RK criterion at 250 hPa:**** The Rayleigh-Kuo stability criterion provides a necessary condition for barotropic and baroclinic instability. Computed as $q/y = -\frac{1}{2} u^2/y^2$, where negative values indicate regions satisfying the instability criterion. The 250 hPa level is chosen as representative of the jet stream, where barotropic processes are strongest.
- ****KE advection at 250 hPa:**** Kinetic energy advection ($-V \cdot KE$) quantifies the tendency for KE to increase or decrease due to advection within the jet stream. Positive values indicate regions where the flow is accelerating, while negative values indicate deceleration. Computed at 250 hPa to capture jet-level dynamics.

- ****AFC at 250 hPa:**** Ageostrophic Flux Convergence (Orlanski & Katzfey, 1991; Orlanski & Sheldon, 1993) quantifies how ageostrophic pressure work redistributes eddy kinetic energy. A ****temporal decomposition**** is used: the 30-year monthly climatology (1991-2020, WMO standard) serves as the base state (V_m , Φ_m), and the instantaneous departure is the eddy perturbation. This is deliberately independent of the area-mean decomposition used in the Lorenz Energy Cycle analysis to avoid circular validation. Positive AFC indicates an eddy KE source; negative values indicate a sink.

Pipeline Structure

Steps

Step	Script	Run on	Description
1	`step1_select_ep_tracks.py`	Local/Remote	Select EP1 and EP2 cyclone tracks
2	`step2_download_era5_parallel.py`	**Remote**	Download ERA5 data (parallel, with patch)
2.1	`step2_1_download_era5_monthly_means.py`	**Remote**	Download ERA5 monthly means → 3
2M	`step2_monitor.py`	Local/Remote	**Monitor download progress** (see below)
3	`step3_precompute_composites.py`	**Remote**	Compute field composites (EGR, PV, adv_T)
4	`step4_create_figures.py`	Local	Create EP1 vs EP2 composite figures
5	`step5_update_scientific_notes.py`	Local	Populate SCIENTIFIC_NOTES.md with regional s

Monthly climatology download (`step2\_1\_download\_era5\_monthly\_means.py`)

Downloads 12-month ERA5 climatologies (1991-2020) for all anomaly diagnostics, organized into 1

Group	Levels (hPa)	Variables	Output file
`250hPa`	250	u, v, z	`era5_climatology_250hPa.nc`
`pv200`	175, 200, 225	u, v, t	`era5_climatology_pv200.nc`
`pv850`	825, 850, 875	u, v, t	`era5_climatology_pv850.nc`
`mfd975`	975	u, v, q	`era5_climatology_mfd975.nc`
`slp`	surface	msl	`era5_climatology_slp.nc`

```
```bash
```

```
Download all groups (auto-skips valid existing files)
```

```
python scripts/ep_structure_analysis/step2_1_download_era5_monthly_means.py
```

```
Download specific groups only
```

```
python scripts/ep_structure_analysis/step2_1_download_era5_monthly_means.py --groups pv200 pv850
```

```
Only recompute climatology files from already-downloaded raw data
```

```
python scripts/ep_structure_analysis/step2_1_download_era5_monthly_means.py --clim-only
```

```
Force re-download of specific months (e.g. June and July)
```

```
python scripts/ep_structure_analysis/step2_1_download_era5_monthly_means.py --force-months 6 7
```

The 250hPa group reuses existing era5\_raw\_month{MM}.nc files without re-downloading, preserving backward compatibility.

### Download monitor (step2\_monitor.py)

step2\_monitor.py scans data/era5\_ep\_structure/ and reports completeness at the finest available granularity: one **slot** = one (variable, pressure-level) pair.

Slots per case = 5 pressure vars × 9 levels + 1 SLP = 46 slots total

**Features:** - **Process detection:** Automatically detects if step2\_download\_era5\_parallel.py is running and shows PID, runtime, CPU%, and memory usage (requires psutil) - **Per-variable table:** how many cases have that variable with all 9 levels - **Per-level table:** how many cases have that level with all 5 variables - **Composite check:** whether precomputed\_composites\_ep{1,2}.nc (step 3 output) already exist

*# Install psutil for process detection (optional but recommended)*

```
pip install psutil
```

*# One-shot report (shows current state and exits)*

```
python scripts/ep_structure_analysis/step2_monitor.py
```

*# Live watch while step 2 is running (refresh every 60 s)*

```
python scripts/ep_structure_analysis/step2_monitor.py --watch
```

*# Faster refresh (every 30 s)*

```
python scripts/ep_structure_analysis/step2_monitor.py --watch --interval 30
```

*# No terminal clear - safe for nohup / log capture*

```
python scripts/ep_structure_analysis/step2_monitor.py --watch --no-clear
```

Example output (with download process active):

#### ERA5 EP STRUCTURE - DOWNLOAD MONITOR

Scanned : 2026-02-20 14:30:00 (3.2 s)

Dir : .../data/era5\_ep\_structure

Slots : 46 per case = 5 pressure vars × 9 levels + 1 SLP

DOWNLOAD ACTIVE PID=12345 Runtime=2h 15m 30s CPU=45.2% RAM=512MB

EP1	slots	[		]	5520/20424	(27.0%)	
	cases	[		]	120/444	(27.0%)	(fully complete)
	detail		complete: 120		partial: 150		missing: 174
	disk		45.2 GB				
EP2	slots	[		]	12150/45034	(27.0%)	
	cases	[		]	264/979	(27.0%)	(fully complete)
	detail		complete: 264		partial: 330		missing: 385
	disk		98.5 GB				

...

## Scientific Documentation

**PDF generation** Generate a professional PDF version of SCIENTIFIC\_NOTES.md:

```
python scripts/ep_structure_analysis/generate_scientific_notes_pdf.py
```

**Requirements:** - Pandoc (Markdown → PDF converter) - LaTeX distribution (pdflatex, for PDF rendering)

### Install on macOS:

```
brew install pandoc basicstex
```

### Install on Linux:

```
sudo apt-get install pandoc texlive-latex-base texlive-fonts-recommended texlive-latex-extra
```

**Output:** docs/scientific\_notes\_ep\_structure.pdf

The script: - Automatically checks for pandoc and pdflatex - Tries eisvogel template first (fancy), falls back to basic - Opens the PDF automatically after generation (optional) - Works on macOS, Linux, and Windows

**Code verification tests** Verify that spherical grid spacing and gradient calculations work correctly:

```
Run test suite
import numpy as np
from step3_precompute_composites import compute_spherical_grid_spacing

Test 1: Uniform field → zero gradient
lat = np.linspace(-30, 30, 10)
lon = np.linspace(-30, 30, 10)
dx, dy, lat_2d, lon_2d = compute_spherical_grid_spacing(lat, lon)

T = np.ones_like(lat_2d) # uniform temperature
dT_dx = np.gradient(T, axis=1) / dx
dT_dy = np.gradient(T, axis=0) / dy[:, np.newaxis]

assert np.allclose(dT_dx, 0)
assert np.allclose(dT_dy, 0)
print(" Test 1 passed: uniform field → zero gradient")

Test 2: Reversed latitude coordinates
lat_rev = np.linspace(30, -30, 10) # decreasing (some datasets have this)
dx_rev, dy_rev, _, _ = compute_spherical_grid_spacing(lat_rev, lon)
Should still give correct physical spacing (positive)
assert np.all(dy_rev > 0)
print(" Test 2 passed: reversed latitude → correct spacing")

Test 3: Linear latitude gradient
T_linear = lat_2d.copy() # Temperature = latitude
dT_dy_analytic = 1.0 / (111320.0) # 1°/distance (approximate)
```



```
dT_dy_numeric = np.gradient(T_linear, axis=0) / dy[:, np.newaxis]
Should be approximately constant
assert np.std(dT_dy_numeric) / np.mean(np.abs(dT_dy_numeric)) < 0.1
print(" Test 3 passed: linear gradient → consistent derivative")
```

**Expected typical values** (for sanity checks during processing):

Dagnostic	Typical Range	Units
EGR (250–850 hPa)	0.3 – 1.5	day <sup>1</sup>
PV @ 200 hPa	1 – 5	PVU
PV @ 850 hPa	0.1 – 1.0	PVU
Temperature advection @ 850 hPa	±2 – 5	K h <sup>1</sup>
Moisture flux divergence @ 975 hPa	±5 – 20	g kg <sup>1</sup> s <sup>1</sup>
SLP minimum	950 – 995	hPa

If values fall far outside these ranges, check: - Coordinate orientation (latitude increasing vs. decreasing) - Unit consistency (K vs. °C, Pa vs. hPa) - Domain extent (should be 30° × 30° centered on cyclone)

See SCIENTIFIC\_NOTES.md Section 7 for full quality control details.

## Remote server execution

Steps 2–3 require significant compute/storage and should run on the remote server:

```
On remote server (master.iag.usp.br)
cd /discos-varal/swell/p1-swell/danilocs/paper_energy_patterns

Step 1 (can run locally or remotely)
python scripts/ep_structure_analysis/step1_select_ep_tracks.py

Step 2 - download ERA5 (use nohup for long-running download)
nohup python scripts/ep_structure_analysis/step2_download_era5_parallel.py --jobs 4 &

Monitor download progress in another terminal
python scripts/ep_structure_analysis/step2_monitor.py --watch

Step 3 - precompute composites
nohup python scripts/ep_structure_analysis/step3_precompute_composites.py &
```

## Transfer to local machine

```
Transfer precomputed composites only (~100–300 MB vs ~30–60 GB raw ERA5)
bash scripts/ep_structure_analysis/transfer_guide_scp.sh
```

Or manually:

```
scp -i ~/Documents/Master/id_rsa.danilocs -C \
 danilocs@master.iag.usp.br:/discos-varal/swell/p1-swell/danilocs/paper_energy_patterns/data/
 ./data/era5_ep_structure/
```

## Local execution (after transfer)

*# Step 4 - create figures*

```
python scripts/ep_structure_analysis/step4_create_figures.py
```

*# Step 5 - update scientific notes with regional statistics*

```
python scripts/ep_structure_analysis/step5_update_scientific_notes.py
```

*# Step 5 - with PDF generation (requires pandoc + xelatex)*

```
python scripts/ep_structure_analysis/step5_update_scientific_notes.py --pdf
```

**Step 5 computes:** - Global statistics (mean, std, min, max) - Regional statistics: Full  $30\times 30^\circ$  domain, Central  $15\times 15^\circ$  LEC domain, NW/NE/SW/SE quadrants  
- Populates all {PLACEHOLDER} variables in SCIENTIFIC\_NOTES.md - Optionally generates PDF via pandoc

## ERA5 Variables Downloaded

**Pressure levels (hPa):** 175, 200, 225, 250, 500, 825, 850, 875, 975

Levels | Purpose |

|———|———|| | 250, 850 | EGR layer bounds (vertical wind shear + static stability) | | 175, 200, 225 | PV at 200 hPa (centered finite difference for  $\partial/\partial p$ ) | | 825, 850, 875 | PV at 850 hPa (centered finite difference for  $\partial/\partial p$ ) | | 850 | Temperature advection (u, v, T at 850 hPa) | | 500 | Mid-tropospheric reference for stability profile | | 975 | Low-level moisture flux (u, v, q at 975 hPa) |

**Pressure-level variables:** u, v, t, z (geopotential), q (specific humidity)

**Single-level variables:** msl (mean sea level pressure)

**Domain:**  $30^\circ \times 30^\circ$  centred on cyclone track centre during intensification

## Output

### Data

- data/era5\_ep\_structure/ — raw ERA5 files (remote only) + precomputed composites
- results/ep\_structure/ — case lists, statistics

### Figures

- figures/ep\_structure/ — EP1 vs EP2 composite comparison panels

#### Total-field figures:

- composite\_egr.png — Eady Growth Rate (250–850 hPa)
- composite\_pv200.png — PV at 200 hPa + 250 hPa wind vectors
- composite\_pv850.png — PV at 850 hPa + 850 hPa wind vectors
- composite\_advT850.png — Temperature advection at 850 hPa
- composite\_moisture\_flux.png — Specific humidity + moisture flux divergence at 975 hPa
- composite\_slp.png — Sea level pressure
- composite\_rk\_criterion.png — Rayleigh-Kuo criterion at 250 hPa

- `composite_ke_advection.png` — Kinetic energy advection at 250 hPa
- `composite_afc_250.png` — AFC at 250 hPa (eddy by construction)

**Anomaly figures** (departure from 1991–2020 climatology; require `step2_1` multi-group download):

- `composite_pv200_anom.png` — PV at 200 hPa + 250 hPa wind vectors
- `composite_pv850_anom.png` — PV at 850 hPa + 850 hPa wind vectors
- `composite_advT850_anom.png` — Temperature advection anomaly at 850 hPa
- `composite_moisture_flux_anom.png` — Moisture flux divergence anomaly at 975 hPa
- `composite_ke_advection_anom.png` — KE advection anomaly at 250 hPa
- `composite_slp_anom.png` — SLP anomaly + 850 hPa wind vectors

Each figure shows EP1 (left) vs EP2 (right).  $15^\circ \times 15^\circ$  dashed box marks the LEC domain.

## Logs

- `logs/ep_structure_*.log` — detailed execution logs

## Data Storage

Component	Size	Location
Raw ERA5 (EP1 + EP2)	~30-60 GB	Remote server only
Precomputed composites	~100-300 MB	Local + remote
Figures	~5-15 MB	Local

## Cluster → EP Mapping

From `scripts/exploratory/analyze_ep_characteristics.py`:

Cluster	Energy Pattern	Ck Characteristic
0	EP1	Strong baroclinic and barotropic
2	EP2	Intermediate conversions and strong imports of energy
1	EP3	Day-to-day cyclones

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## Section: scripts/ck\_subterms\_analysis

Source: scripts/ck\_subterms\_analysis/README.md

# Ck Subterms Analysis: Barotropic Energy Conversion Decomposition

## Scientific Motivation

The barotropic conversion term (Ck) represents the conversion of kinetic energy between the zonal mean and eddy components. In the context of extratropical cyclones in the Southwestern Atlantic, Ck quantifies the energy transfer from eddy kinetic energy (Ke) to zonal mean kinetic energy (Kz):

$$Ck = -\overline{u'v'} \frac{\partial \bar{u}}{\partial y} - \overline{u'w'} \frac{\partial \bar{u}}{\partial p} - \overline{v'w'} \frac{\partial \bar{v}}{\partial p}$$

where: -  $u, v, w$  are zonal, meridional, and vertical wind components - Primes (') denote deviations from the area mean (eddy components) - Overbars ( ) denote area averages (mean state)

**Physical Interpretation:** - **Negative Ck:** Energy flows from eddies to mean flow (Ke → Kz) - Typical during cyclone intensification in baroclinic zones - Eddy momentum fluxes decelerate the mean flow - **Positive Ck:** Energy flows from mean flow to eddies (Kz → Ke) - Associated with barotropic instability - Mean flow accelerates the eddies

## Energy Pattern 1 (EP1) Characteristics

EP1 cyclones exhibit the strongest energetic conversions in the Southwestern Atlantic: - **Mean Ck:** -16.48 W/m<sup>2</sup> (strongest barotropic conversion among all patterns) - **Frequency:** 444 cyclones (11.6% of total) - **Analysis scope:** ALL EP1 cyclones (no spatial restriction) - **Vertical structure:** Ck minimum occurs at mid-tropospheric levels (~500-600 hPa)

## Research Questions

1. **What is the relative importance of each Ck subterm during EP1 cyclone intensification?**
  - Horizontal momentum flux:  $-\overline{u'v'} \frac{\partial \bar{u}}{\partial y}$
  - Vertical momentum flux (zonal):  $-\overline{u'w'} \frac{\partial \bar{u}}{\partial p}$
  - Vertical momentum flux (meridional):  $-\overline{v'w'} \frac{\partial \bar{v}}{\partial p}$
2. **How do Ck subterms evolve throughout the cyclone lifecycle?**
  - Comparison across phases: incipient → intensification → mature → decay
  - Identification of dominant mechanisms in each phase
3. **Are there differences in Ck subterm contributions between different EP1 cyclones?**
  - Regional variations (genesis location, track orientation)
  - Seasonal variations (winter vs. summer systems)
4. **How does the updated LorenzCycleToolkit decomposition compare with previous results?**

- Validation against Zenodo dataset (aggregated Ck)
- Assessment of term-by-term contributions to total Ck

## Methodology

### Prerequisites

**IMPORTANT:** This analysis requires the cluster analysis results: - `results/cluster/kmeans_clustered_data.` (produced by `scripts/cluster_analysis_energy_patterns/`) - This file contains the EP assignments for all cyclones, including all 444 EP1 cyclones - **No spatial restriction** — all EP1 cyclones regardless of genesis location are included

### Workflow Overview

Step 1: Prepare Track Files

↓

Step 2: Run LorenzCycleToolkit (automated ERA5 download & LEC computation)

↓

Step 3: Extract and Analyze Ck Subterms

↓

Step 4: Compare Lifecycle Evolution

↓

Step 5: Statistical Analysis and Visualization

---

### Step 1: Prepare Track Files for LorenzCycleToolkit

**Objective:** Convert ALL EP1 cyclone tracks to LorenzCycleToolkit input format.

**Input:** - `results/cluster/kmeans_clustered_data.csv` — EP assignments from cluster analysis (444 EP1 cyclones) - Main track database (via `load_tracks()`)

**Output Format** (one file per cyclone):

```
time;Lat;Lon
2005-08-08-0000;-22.5;-45.0
2005-08-08-0600;-23.0;-44.5
2005-08-08-1200;-23.5;-44.0
...
```

**File Naming:** `track_{track_id}.txt` - Example: `track_20070643.txt`

**Output Directory:** `data/ck_analysis/tracks/`

**Key Considerations:** - Time format: YYYY-MM-DD-HHMM (no spaces, dash before time) - Header: `time;Lat;Lon` (capitalized) - Coordinates: Latitude and Longitude in decimal degrees - Delimiter: Semicolon (;) with NO spaces - Data source: Full cyclone lifecycle (all phases, all timesteps) - Temporal resolution: 3-hourly (matches project configuration)

**Script:** `step1_prepare_tracks.py`

---

## Step 2: Run LorenzCycleToolkit with ERA5 Download

**Objective:** Compute Lorenz Energy Cycle with full term decomposition using the updated toolkit version.

### LorenzCycleToolkit Configuration:

```
Example configuration (to be adjusted)
from LorenzCycleToolkit import LorenzCycleAnalysis

Initialize analysis
lec = LorenzCycleAnalysis(
 track_file='data/ck_analysis/tracks/track_20070643.txt',
 dataset='ERA5',
 auto_download=True, # Automatically download ERA5 data
 pressure_levels='standard', # All standard pressure levels
 spatial_filter='moving_box', # Follow cyclone center
 box_size=15, # 15° × 15° domain (consistent with EP1 analysis)
 decompose_terms=True # Enable full term decomposition
)

Run analysis
lec.compute()
lec.save_results('results/ck_analysis/lec_results/')

```

**Key Features of Updated Toolkit:** - **Automatic ERA5 download:** No manual CDS API interaction needed - **Term decomposition:** Separates Ck into individual subterms - **Vertical resolution:** Maintains level-by-level information - **Phase identification:** Uses CycloPhaser for lifecycle detection

### Output Structure (per cyclone):

```
results/ck_analysis/lec_results/{track_id}_ERA5_track/
 periods.csv # Lifecycle phases
 results.csv # Integrated terms (all phases)
 Ck_level.csv # Total Ck by pressure level
 Ck_uv_level.csv # Horizontal momentum flux term
 Ck_uw_level.csv # Vertical momentum flux (zonal) term
 Ck_vw_level.csv # Vertical momentum flux (meridional) term
 Ca_level.csv # Baroclinic conversion (for reference)
 ... # Other energy terms

```

**Computational Requirements:** - **Storage:** ~500 MB per cyclone (ERA5 data + results) - **Time:** ~30-60 min per cyclone (depending on lifecycle length) - **Total for ALL EP1:** ~444 cyclones × 500 MB = ~222 GB - **Processing:** Can be parallelized across cyclones

**Script:** `step2_run_lec_toolkit.py` (wrapper script) - Iterates through prepared tracks - Submits LorenzCycleToolkit jobs - Monitors progress and handles errors

### Step 3: Extract and Analyze Ck Subterms

**Objective:** Load LEC results and extract Ck subterm contributions for all EP1 cyclones.

**Analysis Tasks:**

1. **Load Ck Subterms by Level:**
  - Ck\_uv(p): Horizontal momentum flux contribution
  - Ck\_uw(p): Vertical momentum flux (zonal) contribution
  - Ck\_vw(p): Vertical momentum flux (meridional) contribution
  - Validate:  $Ck_{total} = Ck_{uv} + Ck_{uw} + Ck_{vw}$
2. **Phase-Specific Analysis:**
  - Extract values during intensification phase only
  - Compare with mature and decay phases
  - Compute phase-averaged profiles
3. **Vertical Structure:**
  - Identify pressure level of maximum contribution for each subterm
  - Analyze vertical distribution (surface → upper troposphere)
  - Compare with Ck minimum level from Step 2 of EP1 analysis
4. **Relative Importance Metrics:**
  - Percentage contribution:  $\frac{Ck_{subterm}}{Ck_{total}} \times 100\%$
  - Dominant term identification (largest absolute contribution)
  - Term consistency across cyclones (standard deviation)

**Output:** - results/ck\_analysis/subterms\_by\_cyclone.csv - Individual cyclone subterm values  
- results/ck\_analysis/subterms\_statistics.csv - Ensemble statistics  
- results/ck\_analysis/vertical\_profiles.csv - Mean profiles by phase

**Script:** step3\_extract\_subterms.py

---

### Step 4: Lifecycle Evolution Analysis

**Objective:** Quantify how Ck subterm contributions change throughout the cyclone lifecycle.

**Methodology:**

1. **Temporal Evolution:**
  - Time series of each subterm throughout lifecycle
  - Smoothed profiles (24h running mean)
  - Identification of transition points between phases
2. **Phase Composites:**
  - Mean subterm values by phase (incipient, intensification, mature, decay)
  - Vertical structure evolution (2D: pressure × phase)
  - Statistical significance testing (phase-to-phase changes)
3. **Energetic Pathway Analysis:**
  - Dominant mechanism identification by phase
  - Transitions: Which subterm leads intensification onset?
  - Which subterm sustains mature phase?



**Visualization:** - Time-height cross-sections (pressure  $\times$  time) - Phase-composite boxplots (subterm  $\times$  phase) - Stacked area charts (relative contribution evolution)

**Output:** - figures/ck\_analysis/lifecycle\_evolution.png - figures/ck\_analysis/phase\_composites.png  
- results/ck\_analysis/lifecycle\_statistics.csv

**Script:** step4\_lifecycle\_analysis.py

---

## Step 5: Statistical Analysis and Validation

**Objective:** Validate results against previous analysis and identify patterns.

**Validation:**

### 1. Comparison with Zenodo Dataset:

- Compare aggregated Ck from new analysis with old results
- Expected: Close agreement in total Ck values
- Difference: New analysis provides subterm breakdown

### 2. Consistency Checks:

- Mass budget conservation (total energy should be conserved)
- Term magnitude reasonableness (compare with literature values)
- Spatial averaging consistency

**Statistical Tests:**

### 1. Inter-cyclone Variability:

- Standard deviation of subterm contributions
- Coefficient of variation ( $CV = \sigma / \mu$ )
- Identification of outlier cyclones

### 2. Regional/Seasonal Patterns:

- Subterm contributions vs. genesis latitude
- Summer vs. winter differences (if data sufficient)
- Track orientation effects (NE vs. SE propagation)

### 3. Correlation Analysis:

- Subterm vs. cyclone intensity (maximum vorticity)
- Subterm vs. lifecycle duration
- Cross-correlations between subterms

**Output:** - figures/ck\_analysis/validation\_plots.png - figures/ck\_analysis/statistical\_summary.png  
- results/ck\_analysis/final\_report.txt

**Script:** step5\_statistical\_analysis.py

---

## Expected Results

### Hypothesis 1: Horizontal Momentum Flux Dominance

**Expectation:** The horizontal momentum flux term ( $-\overline{u'v'} \frac{\partial \bar{u}}{\partial y}$ ) will dominate Ck during intensification.

**Rationale:** - EP1 cyclones occur in regions of strong baroclinicity - Associated with strong meridional temperature gradients - Thermal wind balance implies strong vertical wind shear - Eddy momentum fluxes ( $u'v'$ ) transport momentum meridionally - This is the “classic” barotropic conversion mechanism

**Expected Magnitude:** ~70-80% of total Ck

---

## **Hypothesis 2: Vertical Level Structure**

**Expectation:** Maximum Ck contributions occur at mid-tropospheric levels (500-700 hPa).

**Rationale:** - Previous analysis shows Ck minimum at ~500-600 hPa - This level corresponds to: - Maximum meridional wind shear ( $\partial u / \partial y$ ) - Jet stream level in South Atlantic - Level of maximum eddy kinetic energy

**Implication:** Barotropic conversion is primarily a mid-tropospheric process in EP1 cyclones

---

## **Hypothesis 3: Phase-Dependent Mechanisms**

**Expectation:** Different subterms dominate in different lifecycle phases.

### **Phase-Specific Predictions:**

#### **1. Incipient Phase:**

- Weak Ck (small absolute values)
- All subterms comparable in magnitude
- Horizontal flux begins to dominate

#### **2. Intensification Phase:**

- Horizontal momentum flux ( $u'v'$ ) dominates
- Strong negative Ck (energy from eddies to mean flow)
- Peak subterm magnitudes

#### **3. Mature Phase:**

- Horizontal flux remains dominant but weakens
- Vertical flux terms become relatively more important
- Beginning of transition to decay

#### **4. Decay Phase:**

- Weakening of all subterms
  - Possible reversal of some terms (positive contributions)
  - Dominance structure may shift
- 

## **Technical Notes**

### **Data Sources**

#### **1. Cyclone Selection:**

- From `results/cluster/kmeans_clustered_data.csv` (cluster analysis results)
- 444 EP1 cyclones with complete lifecycle

- No spatial domain restriction (all EP1 cyclones regardless of genesis location)
2. **ERA5 Reanalysis:**
    - Automatically downloaded by LorenzCycleToolkit
    - Standard pressure levels (1000-100 hPa)
    - 3-hourly temporal resolution
    - 0.25° spatial resolution
  3. **Track Data:**
    - From main database via `load_tracks()`
    - Full lifecycle coverage
    - Vorticity center tracking

## LorenzCycleToolkit Updates

**New Version Features** (compared to Zenodo dataset):

1. **Term Decomposition:**
  - Previous: Only total Ck computed
  - New: Ck separated into three subterms
2. **Pressure Level Output:**
  - Previous: Single `Ck_level.csv` file
  - New: Separate files for each subterm (`Ck_uv_level.csv`, etc.)
3. **Corrections Applied:**
  - Previous version had sign/normalization issues (documented in EP1 analysis)
  - New version: Corrections already implemented
  - No post-processing corrections needed
4. **Automatic Workflow:**
  - Previous: Manual ERA5 download via CDS API
  - New: Integrated download within toolkit
  - Simplified workflow, fewer error points

## Spatial Domain Considerations

**Analysis Domain:**  $15^\circ \times 15^\circ$  box centered on cyclone

**Rationale:** - Consistent with EP1 instability analysis (`step3_download_era5.py`) - Large enough to capture synoptic-scale features - Small enough to focus on cyclone-related processes - Captures eddy momentum fluxes and mean state gradients

**Alternative Domains** (for sensitivity analysis): -  $10^\circ \times 10^\circ$ : More local, emphasizes cyclone core  
 -  $20^\circ \times 20^\circ$ : More regional, includes larger-scale environment

---

## Directory Structure

```
data/ck_analysis/
 tracks/ # Prepared track files
 track_19790001.txt
 track_19790006.txt
 ...
 era5/ # ERA5 data (downloaded by toolkit)
```

```

 {track_id}/
 ...

results/ck_analysis/
 lec_results/ # LorenzCycleToolkit output
 {track_id}_ERA5_track/
 periods.csv
 results.csv
 Ck_level.csv
 Ck_uv_level.csv
 Ck_uw_level.csv
 Ck_vw_level.csv
 subterms_by_cyclone.csv # Individual cyclone data
 subterms_statistics.csv # Ensemble statistics
 vertical_profiles.csv # Mean vertical profiles
 lifecycle_statistics.csv # Phase-composite statistics
 final_report.txt # Summary of findings

figures/ck_analysis/
 vertical_profiles.png # Mean profiles by subterm
 lifecycle_evolution.png # Time evolution composites
 phase_composites.png # Boxplots by phase
 validation_plots.png # Comparison with Zenodo data
 statistical_summary.png # Scatter plots and correlations

```

---

## References

### Lorenz Energy Cycle

- **Lorenz, E. N.** (1955). Available potential energy and the maintenance of the general circulation. *Tellus*, 7(2), 157-167.
- **Brennan, F. E., & Vincent, D. G.** (1980). Zonal and eddy components of the synoptic-scale energy budget during intensification of Hurricane Carmen (1974). *Monthly Weather Review*, 108(7), 954-965.

### Barotropic Conversion

- **Orlanski, I., & Sheldon, J. P.** (1995). Stages in the energetics of baroclinic systems. *Tellus A*, 47(5), 605-628.
- **Dias Pinto, J. R., & da Rocha, R. P.** (2011). The energy cycle and structural evolution of cyclones over southeastern South America in three case studies. *Journal of Geophysical Research*, 116, D14104.

### South Atlantic Cyclones

- **Gan, M. A., & Rao, V. B.** (1991). Surface cyclogenesis over South America. *Monthly Weather Review*, 119(5), 1293-1302.

- **Reboita, M. S., et al.** (2010). Climatology of cyclones over the South Atlantic. *International Journal of Climatology*, 30(11), 1781-1798.
- 

## Future Work

### Sensitivity Analyses

1. **Domain Size Sensitivity:**
  - Repeat analysis with  $10^{\circ} \times 10^{\circ}$  and  $20^{\circ} \times 20^{\circ}$  domains
  - Assess impact on subterm contributions
2. **Vertical Resolution:**
  - Use all available pressure levels vs. subset
  - Impact on vertical derivative calculations
3. **Temporal Resolution:**
  - Effect of 3-hourly vs. 6-hourly data
  - Sub-daily variability in subterm contributions

### Extended Analyses

1. **Comparison with Other Energy Patterns:**
    - EP2, EP3, EP4 cyclones
    - Identify unique features of EP1 barotropic conversion
  2. **Seasonal Variability:**
    - Winter (JJA) vs. summer (DJF) differences
    - Connection to ENSO phases
  3. **Trajectory Analysis:**
    - Subterm evolution along cyclone track
    - Spatial patterns of energy conversion
  4. **Ageostrophic Effects:**
    - Role of ageostrophic circulation in Ck subterms
    - Connection to frontal dynamics
- 

## Section: scripts/preprocess\_data

Source: `scripts/preprocess_data/README.md`

## Preprocess Data

Scripts for downloading input datasets from Zenodo and preprocessing them into formats optimised for analysis scripts.

---

## Scripts

### `download_lec_from_zenodo.py`

Downloads the complete Lorenz Energy Cycle (LEC) results dataset from Zenodo and extracts it locally.

- **Source DOI:** 10.5281/zenodo.18243447
- **Contents:** Complete LEC results with vertical resolution (~1,500 cyclones, 2020, 32 pressure levels, 3-hourly)1979
- **Output:** `data/temp_lec_zenodo/LEC_Results_energetic-patterns/` (one subdirectory per cyclone)
- **Notes:** Checks for existing data and skips re-downloading. Archive is ~2 GB; extraction creates ~6,700 subdirectories.1

### `extract_tracks_from_zenodo.py`

Downloads the integrated cyclone tracks and energetics CSV from Zenodo and writes a smaller, processed version for fast local reads.

- **Source DOI:** 10.5281/zenodo.18133432
- **Output:** `data/tracks_SAt_filtered_with_energetics_processed.csv`
- **Notes:** Subsets to the columns used by plotting scripts to reduce file size.

### `preprocess_data.py`

Loads all cyclone energy data and serialises it to a Parquet cache for 1000 faster access by analysis scripts.

- **Input:** Remote GitHub CSV (accessed via `scripts/utils/load_data.py`)
- **Output:** `data/energy_cache.parquet` (~100 MB)50
- **Notes:** Supports parallel loading (`N_WORKERS` configurable in the script header). Run once; re-run only when the upstream data changes.

### `run_all.py`

Runs all three scripts above in alphabetical order. Prints progress and reports failures.

---

## Run Order

*# Run everything at once (recommended)*

```
python scripts/preprocess_data/run_all.py
```

*# Or run individually in this order:*

```
python scripts/preprocess_data/download_lec_from_zenodo.py
```

```
python scripts/preprocess_data/extract_tracks_from_zenodo.py
```

```
python scripts/preprocess_data/preprocess_data.py
```

---

## Outputs Summary

File	Produced by	Used by
data/temp_lec_zenodo/INTCoReResultsFromEnergyPatterns/	ck_subterms_analysis/	
data/trackSAt_files/extract_tracks_from_csv_processed.csv	scripts/main/, cluster_analysis_energy_patterns/	
data/energy_cache.parenetcess_data.py	scripts/main/, cluster_analysis_energy_patterns/, scripts/exploratory/	